

New 0inflated models: Poisson & Binomial

Parametrisation

This is a new implementation (Nov'22) of zero-inflated Poisson and Binomial likelihood, where we will allow for a linear predictor in both the zero-inflation and in the mean, but one of them needs to consist of fixed effects only. This means the setup will be somewhat different than for other likelihood models.

Details

The zero-inflated likelihood $f_0(y|\dots)$ is defined as

$$f_0(y|\eta_1, \eta_2) = p(\eta_1)1_{[y=0]} + (1 - p(\eta_1))f(y|\eta_2)$$

where $f(y|\dots)$ is either Poisson or Binomial. We allow for two linear predictors in the model, but one needs to be “simple” (i.e. only consists of fixed effects). The other is general and defined via the formula. Normally, the zero-inflation probability is simpler (`family="0..."`)

$$\eta_1 = \text{simple} \quad \eta_2 = \text{formula}$$

but they can also be swapped (`family="0...S"`)

$$\eta_1 = \text{formula} \quad \eta_2 = \text{simple}$$

Link-function

This is similar to Poisson and Binomial.

The link-function for the 'simple'-model must be given by argument `link.simple` in the `control.family`-argument. Only link-models without covariates/parameters are currently available. The examples later on show how this is done.

Hyperparameters

All parameters in the simple model are treated as hyperparameters. The j 'th element of η_1 is

$$(\eta_1)_j = \sum_{i=1}^m \beta_i x_{ij}$$

for covariates $x_1, \dots$, where m is maximum 10. An intercept in this model has to be defined manually by adding a constant covariate vector.

New 1inflated models for Poisson

There are two similar models for the Poisson based on the “conditioned to be positive” Poisson $f(y|y > 0)$. These ones are named `"1poisson"` and `"1poissonS"` and are 1-inflated.

Specification

- `family="0poisson"`
- `family="0poissonS"`
- `family="0binomial"`
- `family="0binomialS"`

- `family="1poisson"`
- `family="1poissonS"`
- Required arguments: As for the Poisson and Binomial (but how these arguments are given, will differ). Optional argument `link.simple`.

Hyperparameter specification and default values

Opoisson

doc New 0-inflated Poisson

hyper

theta1

hyperid 56201
name beta1
short.name beta1
output.name beta1 for Opoisson observations
output.name.intern beta1 for Opoisson observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56202
name beta2
short.name beta2
output.name beta2 for Opoisson observations
output.name.intern beta2 for Opoisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56203
name beta3
short.name beta3
output.name beta3 for Opoisson observations
output.name.intern beta3 for Opoisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56204
name beta4
short.name beta4
output.name beta4 for Opoisson observations
output.name.intern beta4 for Opoisson observations
initial 0
fixed FALSE

```

    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56205
    name beta5
    short.name beta5
    output.name beta5 for Opoisson observations
    output.name.intern beta5 for Opoisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56206
    name beta6
    short.name beta6
    output.name beta6 for Opoisson observations
    output.name.intern beta6 for Opoisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56207
    name beta7
    short.name beta7
    output.name beta7 for Opoisson observations
    output.name.intern beta7 for Opoisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56208
    name beta8
    short.name beta8
    output.name beta8 for Opoisson observations
    output.name.intern beta8 for Opoisson observations
    initial 0
    fixed FALSE
    prior normal

```

```

param 0 10
to.theta function(x) x
from.theta function(x) x
theta9
  hyperid 56209
  name beta9
  short.name beta9
  output.name beta9 for Opoisson observations
  output.name.intern beta9 for Opoisson observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56210
  name beta10
  short.name beta10
  output.name beta10 for Opoisson observations
  output.name.intern beta10 for Opoisson observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
survival FALSE
discrete TRUE
link default log quantile
link.simple default logit cauchit probit cloglog ccloglog
pdf 0inflated

```

OpoissonS

doc New 0-inflated Poisson Swap

hyper

theta1

hyperid 56301
name beta1
short.name beta1
output.name beta1 for OpoissonS observations
output.name.intern beta1 for OpoissonS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56302
name beta2
short.name beta2
output.name beta2 for OpoissonS observations
output.name.intern beta2 for OpoissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56303
name beta3
short.name beta3
output.name beta3 for OpoissonS observations
output.name.intern beta3 for OpoissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56304
name beta4
short.name beta4
output.name beta4 for OpoissonS observations
output.name.intern beta4 for OpoissonS observations
initial 0
fixed FALSE
prior normal

```

    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56305
    name beta5
    short.name beta5
    output.name beta5 for OpoissonS observations
    output.name.intern beta5 for OpoissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56306
    name beta6
    short.name beta6
    output.name beta6 for OpoissonS observations
    output.name.intern beta6 for OpoissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56307
    name beta7
    short.name beta7
    output.name beta7 for OpoissonS observations
    output.name.intern beta7 for OpoissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56308
    name beta8
    short.name beta8
    output.name beta8 for OpoissonS observations
    output.name.intern beta8 for OpoissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10

```

```

    to.theta function(x) x
    from.theta function(x) x
theta9
  hyperid 56309
  name beta9
  short.name beta9
  output.name beta9 for OpoissonS observations
  output.name.intern beta9 for OpoissonS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56310
  name beta10
  short.name beta10
  output.name beta10 for OpoissonS observations
  output.name.intern beta10 for OpoissonS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x

survival FALSE
discrete TRUE
link default logit loga cauchit probit cloglog ccloglog loglog log sslogit logitoffset quantile pquantile
link.simple default log
pdf 0inflated

```


0binomial

doc New 0-inflated Binomial

hyper

theta1

hyperid 56401
name beta1
short.name beta1
output.name beta1 for 0binomial observations
output.name.intern beta1 for 0binomial observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56402
name beta2
short.name beta2
output.name beta2 for 0binomial observations
output.name.intern beta2 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56403
name beta3
short.name beta3
output.name beta3 for 0binomial observations
output.name.intern beta3 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56404
name beta4
short.name beta4
output.name beta4 for 0binomial observations
output.name.intern beta4 for 0binomial observations
initial 0
fixed FALSE
prior normal

```

    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56405
    name beta5
    short.name beta5
    output.name beta5 for 0binomial observations
    output.name.intern beta5 for 0binomial observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56406
    name beta6
    short.name beta6
    output.name beta6 for 0binomial observations
    output.name.intern beta6 for 0binomial observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56407
    name beta7
    short.name beta7
    output.name beta7 for 0binomial observations
    output.name.intern beta7 for 0binomial observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56408
    name beta8
    short.name beta8
    output.name beta8 for 0binomial observations
    output.name.intern beta8 for 0binomial observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10

```

```

    to.theta function(x) x
    from.theta function(x) x
theta9
  hyperid 56409
  name beta9
  short.name beta9
  output.name beta9 for 0binomial observations
  output.name.intern beta9 for 0binomial observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56410
  name beta10
  short.name beta10
  output.name beta10 for 0binomial observations
  output.name.intern beta10 for 0binomial observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x

survival FALSE
discrete TRUE
link default logit loga cauchit probit cloglog ccloglog loglog log
link.simple default logit cauchit probit cloglog ccloglog
pdf 0inflated

```

0poisson

doc New 0-inflated Binomial Swap

hyper

theta1

hyperid 56501
name beta1
short.name beta1
output.name beta1 for 0binomialS observations
output.name.intern beta1 for 0binomialS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56502
name beta2
short.name beta2
output.name beta2 for 0binomialS observations
output.name.intern beta2 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56503
name beta3
short.name beta3
output.name beta3 for 0binomialS observations
output.name.intern beta3 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56504
name beta4
short.name beta4
output.name beta4 for 0binomialS observations
output.name.intern beta4 for 0binomialS observations
initial 0
fixed FALSE
prior normal

```

    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56505
    name beta5
    short.name beta5
    output.name beta5 for 0binomialS observations
    output.name.intern beta5 for 0binomialS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56506
    name beta6
    short.name beta6
    output.name beta6 for 0binomialS observations
    output.name.intern beta6 for 0binomialS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56507
    name beta7
    short.name beta7
    output.name beta7 for 0binomialS observations
    output.name.intern beta7 for 0binomialS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56508
    name beta8
    short.name beta8
    output.name beta8 for 0binomialS observations
    output.name.intern beta8 for 0binomialS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10

```

```

    to.theta function(x) x
    from.theta function(x) x
theta9
  hyperid 56509
  name beta9
  short.name beta9
  output.name beta9 for 0binomialS observations
  output.name.intern beta9 for 0binomialS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56510
  name beta10
  short.name beta10
  output.name beta10 for 0binomialS observations
  output.name.intern beta10 for 0binomialS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x

survival FALSE
discrete TRUE
link default logit loga cauchit probit cloglog ccloglog loglog log
link.simple default logit cauchit probit cloglog ccloglog
pdf 0inflated

```

1poisson

doc New 1-inflated Poisson

hyper

theta1

hyperid 56401
name beta1
short.name beta1
output.name beta1 for 1poisson observations
output.name.intern beta1 for 1poisson observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56402
name beta2
short.name beta2
output.name beta2 for 1poisson observations
output.name.intern beta2 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56403
name beta3
short.name beta3
output.name beta3 for 1poisson observations
output.name.intern beta3 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56404
name beta4
short.name beta4
output.name beta4 for 1poisson observations
output.name.intern beta4 for 1poisson observations
initial 0
fixed FALSE
prior normal

```

    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56405
    name beta5
    short.name beta5
    output.name beta5 for 1poisson observations
    output.name.intern beta5 for 1poisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56406
    name beta6
    short.name beta6
    output.name beta6 for 1poisson observations
    output.name.intern beta6 for 1poisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56407
    name beta7
    short.name beta7
    output.name beta7 for 1poisson observations
    output.name.intern beta7 for 1poisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56408
    name beta8
    short.name beta8
    output.name beta8 for 1poisson observations
    output.name.intern beta8 for 1poisson observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10

```



```

    to.theta function(x) x
    from.theta function(x) x
theta9
  hyperid 56409
  name beta9
  short.name beta9
  output.name beta9 for 1poisson observations
  output.name.intern beta9 for 1poisson observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56410
  name beta10
  short.name beta10
  output.name beta10 for 1poisson observations
  output.name.intern beta10 for 1poisson observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x

survival FALSE
discrete TRUE
link default log quantile
link.simple default logit cauchit probit cloglog ccloglog
pdf 0inflated

```

1poissonS

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hyper

theta1

hyperid 56501
name beta1
short.name beta1
output.name beta1 for 1poissonS observations
output.name.intern beta1 for 1poissonS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56502
name beta2
short.name beta2
output.name beta2 for 1poissonS observations
output.name.intern beta2 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56503
name beta3
short.name beta3
output.name beta3 for 1poissonS observations
output.name.intern beta3 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56504
name beta4
short.name beta4
output.name beta4 for 1poissonS observations
output.name.intern beta4 for 1poissonS observations
initial 0
fixed FALSE
prior normal

```

    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta5
    hyperid 56505
    name beta5
    short.name beta5
    output.name beta5 for 1poissonS observations
    output.name.intern beta5 for 1poissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta6
    hyperid 56506
    name beta6
    short.name beta6
    output.name beta6 for 1poissonS observations
    output.name.intern beta6 for 1poissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta7
    hyperid 56507
    name beta7
    short.name beta7
    output.name beta7 for 1poissonS observations
    output.name.intern beta7 for 1poissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10
    to.theta function(x) x
    from.theta function(x) x
theta8
    hyperid 56508
    name beta8
    short.name beta8
    output.name beta8 for 1poissonS observations
    output.name.intern beta8 for 1poissonS observations
    initial 0
    fixed FALSE
    prior normal
    param 0 10

```

```

    to.theta function(x) x
    from.theta function(x) x
theta9
  hyperid 56509
  name beta9
  short.name beta9
  output.name beta9 for 1poissonS observations
  output.name.intern beta9 for 1poissonS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x
theta10
  hyperid 56510
  name beta10
  short.name beta10
  output.name beta10 for 1poissonS observations
  output.name.intern beta10 for 1poissonS observations
  initial 0
  fixed FALSE
  prior normal
  param 0 10
  to.theta function(x) x
  from.theta function(x) x

survival FALSE
discrete TRUE
link default logit loga cauchit probit cloglog ccloglog loglog log sslogit logitoffset quantile pquantile
link.simple default log
pdf 0inflated

```

Example: 0Poisson

```
sim.poisson <- function(prob, m)
{
  stopifnot(length(prob) == length(m) && length(prob) > 0)
  n <- length(m)
  y <- numeric(n)
  event <- (runif(n) < prob)
  idx.zero <- which(event)
  idx.non.zero <- which(!event)
  y[idx.zero] <- 0
  y[idx.non.zero] <- rpois(length(idx.non.zero), lambda = m[idx.non.zero])
  return (y)
}

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

n <- 1000
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
E <- runif(n, min = 0.8, max = 1/0.8)

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
m <- E*exp(eta2)

ok <- FALSE
while(!ok) {
  y <- sim.poisson(prob, m)
  ok <- !all(y == 0)
}

## head(data.frame(y, E, x, z, xx, zz))

r <- inla(
  inla.mdata(cbind(y, E), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
  family = "0poisson",
  data = data.frame(y, E, x, z, xx, zz),
  control.fixed = list(prec = 1, prec.intercept = 1),
  control.compute = list(cpo = TRUE),
  control.family = list(link.simple = link.simple,
    hyper = list(beta1 = list(param = c(0, 1)),
      beta2 = list(param = c(0, 1)),
      beta3 = list(param = c(0, 1)),
      beta4 = list(param = c(0, 1)),
      beta5 = list(param = c(0, 1))))))

rr <- inla(
```

```

inla.mdata(cbind(y, E), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
family = "0poissonS",
data = data.frame(y, E, x, z, xx, zz),
control.fixed = list(prec = 1, prec.intercept = 1),
control.compute = list(cpo = TRUE),
## in this case we need to define link.simple as the main link
control.family = list(control.link = list(model = link.simple),
                      hyper = list(beta1 = list(param = c(0, 1)),
                                   beta2 = list(param = c(0, 1)),
                                   beta3 = list(param = c(0, 1)),
                                   beta4 = list(param = c(0, 1)),
                                   beta5 = list(param = c(0, 1)))))

summary(r)
summary(rr)

res <- cbind("beta" = beta,
            "0poisson" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
            "0poissonS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
            diff = (res[, 2]-beta),
            diffS = (res[, 3]-beta),
            "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
            "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".poisson"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))

```

Example: Binomial

```
sim.binomial <- function(prob, p, size)
{
  ## - prob=zero-inflation-prob
  ## - binomial(size, p)
  stopifnot(length(prob) == length(p) && length(prob) == length(size)
            && length(prob) > 0)
  n <- length(prob)
  y <- numeric(n)
  event <- (runif(n) < prob)
  idx.zero <- which(event)
  idx.non.zero <- which(!event)
  y[idx.zero] <- 0
  y[idx.non.zero] <- rbinom(length(idx.non.zero),
                           size = size[idx.non.zero],
                           prob = p[idx.non.zero])

  return (y)
}

n <- 1000
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
Ntrials <- sample(1:10, n, replace = TRUE)

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
p <- 1/(1 + exp(-eta2))

ok <- FALSE
while(!ok) {
  y <- sim.binomial(prob, p, Ntrials)
  ok <- !all(y == 0)
}

## head(data.frame(y, Ntrials, x, z, xx, zz))

r <- inla(
  inla.mdata(cbind(y, Ntrials), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
  family = "0binomial",
  data = data.frame(y, Ntrials, x, z, xx, zz),
  control.fixed = list(prec = 1, prec.intercept = 1),
  control.compute = list(cpo = TRUE),
  control.family = list(link.simple = link.simple,
                        hyper = list(beta1 = list(param = c(0, 1)),
                                      beta2 = list(param = c(0, 1)),
```

```

beta3 = list(param = c(0, 1)),
beta4 = list(param = c(0, 1)),
beta5 = list(param = c(0, 1))))

rr <- inla(
  inla.mdata(cbind(y, Ntrials), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
  family = "0binomialS",
  data = data.frame(y, Ntrials, x, z, xx, zz),
  control.fixed = list(prec = 1, prec.intercept = 1),
  control.compute = list(cpo = TRUE),
  ## in this case we need to define link.simple as the main link
  control.family = list(control.link = list(model = link.simple),
    hyper = list(beta1 = list(param = c(0, 1)),
      beta2 = list(param = c(0, 1)),
      beta3 = list(param = c(0, 1)),
      beta4 = list(param = c(0, 1)),
      beta5 = list(param = c(0, 1)))))

summary(r)
summary(rr)

res <- cbind("beta" = beta,
  "0binomial" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
  "0binomialS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
  diff = (res[, 2]-beta),
  diffS = (res[, 3]-beta),
  "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
  "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".binomial"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))

```


Example: 1Poisson

```
sim.1poisson <- function(prob, lambda) {
  stopifnot(length(prob) == length(lambda) && length(prob) > 0)
  n <- length(lambda)
  y <- numeric(n)
  event <- (runif(n) < prob)
  idx.one <- which(event)
  idx.non.one <- which(!event)
  y[idx.one] <- 1
  for(i in idx.non.one) {
    y[i] <- 0
    while(y[i] == 0) {
      yy <- rpois(20, lambda = lambda[i])
      if (any(yy > 0)) {
        y[i] <- yy[min(which(yy > 0))]
      }
    }
  }
  return (y)
}

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

n <- 10^4
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
E <- runif(n, min = 0.1, max = 100)

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
lambda <- E*exp(eta2)

y <- sim.1poisson(prob, lambda)

## head(data.frame(y, E, x, z, xx, zz))

r <- inla(
  inla.mdata(cbind(y, E), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
  family = "1poisson",
  data = data.frame(y, E, x, z, xx, zz),
  control.fixed = list(prec = 1, prec.intercept = 1),
  control.compute = list(cpo = TRUE),
  control.family = list(link.simple = link.simple,
    hyper = list(beta1 = list(param = c(0, 1)),
      beta2 = list(param = c(0, 1)),
      beta3 = list(param = c(0, 1)),
      beta4 = list(param = c(0, 1)),
```

```

beta5 = list(param = c(0, 1))))

rr <- inla(
  inla.mdata(cbind(y, E), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
  family = "lpoissonS",
  data = data.frame(y, E, x, z, xx, zz),
  control.fixed = list(prec = 1, prec.intercept = 1),
  control.compute = list(cpo = TRUE),
  control.inla = list(cmin = 0),
  ## in this case we need to define link.simple as the main link
  control.family = list(control.link = list(model = link.simple),
    hyper = list(beta1 = list(param = c(0, 1)),
      beta2 = list(param = c(0, 1)),
      beta3 = list(param = c(0, 1)),
      beta4 = list(param = c(0, 1)),
      beta5 = list(param = c(0, 1))))))

summary(r)
summary(rr)

res <- cbind("beta" = beta,
  "lpoisson" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
  "lpoissonS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
  diff = (res[, 2]-beta),
  diffS = (res[, 3]-beta),
  "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
  "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".poisson"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))

```